

BIL401

PROJECT PROPOSAL

Beril Aydın 231101002

Kaan Baykal 221101028

Real-Time Network Intrusion Detection Pipeline

Description of the project

This project focuses on designing, implementing, and evaluating a highly scalable, fault-tolerant, real-time data streaming pipeline for network intrusion detection. Aligned with the Systems Track requirements, the pipeline simulates high throughput network traffic by replaying historical cyberattack logs as a live data stream. The architecture orchestrates Apache Kafka for message buffering and partition based load distribution, Apache DataFusion for low latency stream processing and periodic anomaly filtering, and an Apache Cassandra cluster for the distributed storage of identified threats and traffic aggregations. Deployed entirely on a locally simulated distributed environment using Docker Compose, this project will evaluate system throughput, network latency, query execution times, and fault tolerance under heavy synthetic loads.

Source of the dataset

The primary dataset is the CICIDS2017 (Canadian Institute for Cybersecurity Intrusion Detection System) dataset. The data will be downloaded as static CSV files and dynamically transformed into a live, high velocity stream to simulate a real time network environment.

- **Source URL:** www.kaggle.com/datasets/chethuhn/network-intrusion-dataset

Description of the dataset

- **Domain:** Cybersecurity, Network Traffic Analysis.
- **Size:** The dataset contains approximately 2.8 million records, resulting in roughly 880 MB of raw data.
- **Key Columns (Features):** The dataset consists of 78 network traffic features. The stream processing pipeline will primarily utilize:
 - **Destination Port:** Utilized as the Kafka partitioning key to distribute the processing load across nodes and test network shuffle performance, as it provides a realistic cardinality for traffic routing.
 - **Injected Timestamp:** Since the preprocessed dataset provides aggregate flow features rather than exact raw timestamps, the data ingestion script will artificially inject real time timestamps into the stream. This is critical for DataFusion to compute sliding and tumbling time windows.
 - **Traffic & Flow Metrics:** Features such as Flow Duration, Destination Port, Total Length of Fwd Packets, and TCP Flags will be used to implement predicate pushdown filtering strategies and threat detection rules.
 - **Label:** The target classification, used to validate a rule based routing and filtering logic in the pipeline.

Platforms, Frameworks and Tools

- **Data Ingestion:** Python to build an asynchronous producer that injects timestamps and replays the CSV data into the pipeline.

- **Message Broker:** Apache Kafka to decouple data generation from processing and handle system backpressure.
- **Processing Engine:** Apache DataFusion. Selected specifically for its memory efficient, Rust based execution model, it will execute complex SQL queries and windowing functions over the streaming data without the heavy JVM overhead associated with traditional Spark deployments.
- **Distributed Storage:** Apache Cassandra (NoSQL). Designed with high write throughput in mind to store system flagged anomalies and windowed metrics.
- **Orchestration & Infrastructure:** Docker Compose. The entire architecture will be containerized to orchestrate a multi node cluster environment locally.
- **Version Control:** Git & GitHub for managing concurrent configuration changes and tracking project evolution.